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Panelboards

Section 7

Panelboards

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ReliaGear® Merchandised neXT SuperBox Power Panelboards

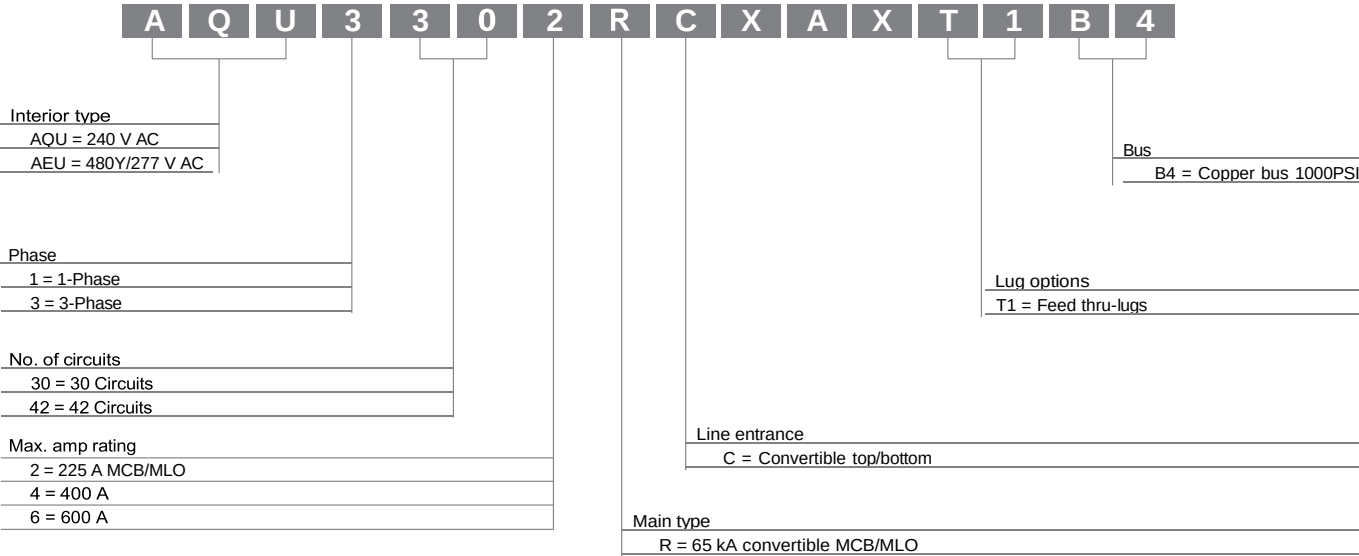
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ReliaGear® Merchandised Lighting Panelboards

Ordering code construction



ReliaGear® Merchandised Lighting Panelboards

Step 1: Select Interior



Select the interior by panel rating, number of circuits and ampacity. Identify the box/front height for interior selected.

Feed-thru copper bus

Voltage	Rating (amps)	No. of circuits	Ordering code ¹	Box/front height (in.)	No. of TGL2 ground bars required ²
240 V AC, 1-Phase	100–225	30	AQU1302RCXAXT1B4	43.5	3
		42	AQU1422RCXAXT1B4	49.5	4
	400	42	AQU1424RCXAXT1B4	76.5	4
		42	AQU1426RCXAXT1B4	76.5	4
208Y/120 V AC, 3-Phase	100–225	30	AQU3302RCXAXT1B4	43.5	3
		42	AQU3422RCXAXT1B4	49.5	4
	400	42	AQU3424RCXAXT1B4	76.5	4
		42	AQU3426RCXAXT1B4	76.5	4
480Y/277 V AC, 3-Phase	100–225	30	AEU3302RCXAXT1B4	43.5	3
		42	AEU3422RCXAXT1B4	49.5	4
	400	42	AEU3424RCXAXT1B4	76.5	4
		42	AEU3426RCXAXT1B4	76.5	4

¹ See table below for TGL20 ground lug quantities.
² For isolated ground, use EGS12. When using the EGS12, 5 and 7 ground lugs (TGL20s) are required for 30 and 42 circuit panels respectively.
³ 600 A available as main lug only.

TGL20 ground lugs required by panel type

Interior type	100-225 A	400 A	600 A
Main lug only	1	1	2
Main lug and sub-feed	2	2	4
Main lug and feed-thru	2	2	4
Main breaker only	1	1	
Main breaker and sub-feed	2	2	
Main breaker and feed-thru	2	2	

ReliaGear® Merchandised Lighting Panelboards

Step 2-3: Select Box and Front



Select a 20" wide box of correct height based on interior selected on Page 7-2.

Note: Boxes come with blank endwalls.

Box height (in.)	NEMA 1 ordering code	NEMA 3R ordering code
43.5	AB43B	AB433
49.5	AB49B	AB493
76.5	AB76B	AB763



For NEMA 1 enclosures only, select a front of correct height based on interior selected on Page 7-2 and box selected above.

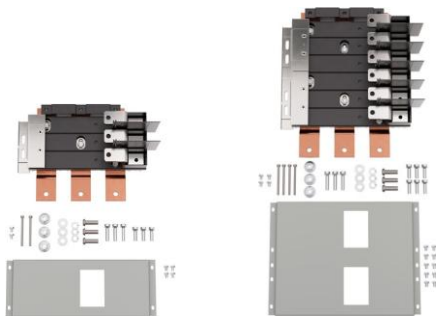
- Standard fronts are equipped with concealed hinges and trim adjusting screws hinges, trim adjusting screws and quarter-turn locks
- Door within door fronts are convenient for easy access to equipment from the front of the panel as it allows access to the gutters without removing the front
- Front hinged to box fronts are similar to door within door fronts for convenient access to gutters, but four screws must be removed to access the outer door

Front height (in.)	Standard	Standard	Door within door	Front hinged to box
	Flush ordering code	Surface ordering code	Surface ordering code	Surface ordering code
43.5	AF43F	AF43S	AF43SP	AF43SD
49.5	AF49F	AF49S	AF49SP	AF49SD
76.5 ¹⁾	AF76FT	AF76ST	AF76SPT	AF76SDT

1) 76.5" fronts include quarter-turn lock, enabling applications with XT5 circuit breakers

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Step 4: Select main and/or sub-feed breaker kit(s)



MBA13

MBA16

Select the main breaker kit appropriate for your interior type, amp rating and kAIC rating. If a sub-feed breaker is required, repeat the selection process.

Interior type	Ordering code ¹	Rating (amps)	No. of poles	Breaker short circuit rating (kAIC)						
				10	14	22	25	35	50	65
AQU1: 240 V AC, 1-phase	MB612	100	2	THQB	—	THHQB	—	—	—	—
	MB614	100	4	(x2) THQB	—	(x2) THHQB	—	—	—	—
	MBA12	225	2	A2A	—	A2N2	—	—	—	—
	MBM324	400	2	—	—	—	—	—	—	XT5N
AQU3: 208Y/120 V AC, 3-phase	MB613	100	3	THQB	—	THHQB	—	—	—	—
	MB616 ⁴	100	6	(x2) THQB	—	(x2) THHQB	—	—	—	—
	MBA13	225	3	A2A	—	A2N ²	—	—	—	—
	MBA16	400	6	(x2) A2A	—	(x2) A2N ²	—	—	—	—
	MBB33	150	3	—	—	—	—	—	—	XT4N
	MBB33	225	3	—	—	—	—	—	—	XT4N
	MBB36 ³	400	6 ³	—	—	—	—	—	—	(x2) XT4N
AEU3: 480Y/277 V AC, 3-phase	MBM334	400	3	—	—	—	—	—	—	XT5N
	MB423	100	3	—	TEY	—	—	—	—	—
	MB426	100	6	—	(x2) TEY	—	—	—	—	—
	MBC33	125	3	—	—	—	XT1N	XT1S	—	XT1H
	MBB33	150	3	—	—	—	XT4N	XT4S	—	XT4H
	MBB33	225	3	—	—	—	XT4N	XT4S	—	XT4H
	MBB36 ³	400	6 ³	—	—	—	(x2) XT4N	(x2) XT4S	—	(x2) XT4H
	MBM334	400	3	—	—	—	—	XT5N	XT5S	XT5H

¹ Breaker not included

² Actual breaker short circuit rating is 25 kAIC

³ 6 poles of sub-feed applies only to 400 A and 600 A interiors

⁴ Can use (2) 3-pole devices only, no 2-pole allowed

ReliaGear® Merchandised Lighting Panelboards

Step 5: Select main and/or sub-feed circuit breaker(s)

To correlate breaker types with the kAIC rating in specific panelboards, see the table for Step 4. For THQB and TEY main breakers, see branch breakers tables in Step 7.

Tmax® XT breakers (3-pole) for use with appropriate main breaker or sub-feed breaker kit.

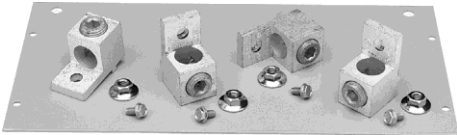
240 V kAIC	480 V kAIC	Breaker description	Wire range (Cu/Al)	Cables per lug	Ordering code
65	65	XT1H 125 TMF 30 amps 3P	Cu Al 1x14–2/0 AWG	1	XT1HU3030AFD000XXX
65	65	XT1H 125 TMF 60 amps 3P	Cu Al 1x14–2/0 AWG	1	XT1HU3060AFD000XXX
65	65	XT1H 125 TMF 100 amps 3P	Cu Al 1x14–2/0 AWG	1	XT1HU3100AFD000XXX
65	65	XT1H 125 TMF 125 amps 3P	Cu Al 1x14–2/0 AWG	1	XT1HU3125AFD000XXX
65	35	XT1S 125 TMF 100 amps 3P	Cu Al 1x14–2/0 AWG	1	XT1SU3100AFD000XXX
65	65	XT4H 250 TMF 150 amps 3P	Cu Al 1x4 AWG–300 kcmil	1	XT4HU3150AFJ000XXX
65	65	XT4H 250 TMF 200 amps 3P	Cu Al 1x4 AWG–300 kcmil	1	XT4HU3200AFJ000XXX
65	65	XT4H 250 TMF 225 amps 3P	Cu Al 1x4 AWG–300 kcmil	1	XT4HU3225AFJ000XXX
65	35	XT4S 250 TMF 175 amps 3P	Cu Al 1x4 AWG–300 kcmil	1	XT4SU3175AFJ000XXX
65	35	XT4S 250 TMF 200 amps 3P	Cu Al 1x4 AWG–300 kcmil	1	XT4SU3200AFJ000XXX
65	35	XT4S 250 TMF 225 amps 3P	Cu Al 1x4 AWG–300 kcmil	1	XT4SU3225AFJ000XXX
65	25	XT4N 250 Ekip DIP 60–150 amps 3P	Cu Al 1x4 AWG–300 kcmil	1	XT4NU31501FJ000XXX
65	35	XT4S 250 Ekip DIP 100–250 amps 3P	Cu Al 1x4 AWG–300 kcmil	1	XT4SU32501FL000XXX
65	65	XT4H 250 Ekip DIP 100–250 amps 3P	Cu Al 1x4 AWG–300 kcmil	1	XT4HU32501FL000XXX
65	65	XT5H 400 TMA 400 amps 3P	Cu Al 2x2/0 AWG–500 kcmi	2	XT5HU340ABFN000XXX
65	50	XT5S 400 TMA 300 amps 3P	Cu Al 2x2/0 AWG–500 kcmi	2	XT5SU330ABFN000XXX
65	50	XT5S 400 TMA 400 amps 3P	Cu Al 2x2/0 AWG–500 kcmi	2	XT5SU340ABFN000XXX
65	35	XT5N 400 Ekip DIP 16–400 amps 3P	Cu Al 2x2/0 AWG–500 kcmi	2	XT5NU340AFFN000XXX
65	65	XT5H 400 Ekip DIP 16–400 amps 3P	Cu Al 2x2/0 AWG–500 kcmi	2	XT5HU340AFFN000XXX

Main or sub-feed breakers for use with RQ panels (208Y/120 V AC 3-phase or 240 V AC 1-phase).

Amp rating	2-pole		3-pole	
	10 kAIC	22 kAIC	10 kAIC	22 kAIC
	Ordering code	Ordering code	Ordering code	Ordering code
125	A2A125TL-2	A2N125TL-2	A2A125TT	A2N125TT
150	A2A150TL-2	A2N150TL-2	A2A150TT	A2N150TT
175	A2A175TL-2	A2N175TL-2	A2A175TT	A2N175TT
200	A2A200TL-2	A2N200TL-2	A2A200TT	A2N200TT
225	A2A225TL-2	A2N225TL-2	A2A225TT	A2N225TT

ReliaGear® Merchandised Lighting Panelboards

Step 6: Select main and/or feed-thru lug kit(s)



Select lug kit(s) for main lug and/or feed-thru applications, if required.

Main lug kits

Lug type	Amp rating	Wire range	Ordering code	200% Neutral ordering code
Mechanical	225	Standard: 6–350 kcmil	MLA1	NKA
	225	Oversized: 1–600 kcmil or (2) 1/0–250 kcmil	MLA2	
	400	Standard: 2–600 kcmil or (2) 1/0–250 kcmil	MLA41	NK4A ¹
	400	Oversized: 3/0–800 kcmil	MLA62	
	600	Standard: (2) 2/0–500 kcmil	MLA61	–
	600	Oversized: 3/0–800 kcmil	MLA62	

¹ For 200% neutral feed-thru, order NKA4FT, (GO-101P). Wire range (2) 2/0–600 kcmil or (4) 4–250 kcmil.

ReliaGear® Merchandised Lighting Panelboards

Step 7: Select bolt-on branch circuit breakers

Select branch circuit breakers from the tables below.

- THQB and THHQB breakers are only compatible with RQ panels.
- TEY and TEYF breakers are only compatible with RE panels.

Branch circuit breakers for use with RQ panels (208Y/120 V AC 3-phase or 240 V AC 1-phase)

Amp rating	10 kAIC			22 kAIC		
	1-pole	2-pole	3-pole	1-pole	2-pole	3-pole
	Ordering code	Ordering code	Ordering code	Ordering code	Ordering code	Ordering code
15	THQB1115 ^{1,2,3,4}	THQB2115 ^{1,4}	THQB32015	THHQB1115 ^{1,2,3,4}	THHQB2115 ^{1,4}	THHQB32015
20	THQB1120 ^{1,2,3,4}	THQB2120 ^{1,4}	THQB32020	THHQB1120 ^{1,2,3,4}	THHQB2120 ^{1,4}	THHQB32020
25	THQB1125 ¹	THQB2125 ¹	THQB32025	THHQB1125 ¹	THHQB2125 ¹	THHQB32025
30	THQB1130 ^{1,4}	THQB2130 ^{1,4}	THQB32030	THHQB1130 ^{1,4}	THHQB2130 ^{1,4}	THHQB32030
35	THQB1135	THQB2135	THQB32035	THHQB1135	THHQB2135	THHQB32035
40	THQB1140	THQB2140 ⁴	THQB32040	THHQB1140	THHQB2140	THHQB32040
45	THQB1145	THQB2145	THQB32045	THHQB1145	THHQB2145	THHQB32045
50	THQB1150	THQB2150	THQB32050	THHQB1150	THHQB2150	THHQB32050
60	THQB1160	THQB2160	THQB32060	THHQB1160	THHQB2160	THHQB32060
70	THQB1170	THQB2170	THQB32070	THHQB1170	THHQB2170	THHQB32070
80	–	THQB2180	THQB32080	–	THHQB2180	THHQB32080
90	–	THQB2190	THQB32090	–	THHQB2190	THHQB32090
100	–	THQB21100	THQB32100	–	THHQB21100	THHQB32100

¹ For ground fault, add 'GFT' suffix to breaker SKU

² For AFCI, add 'AF2' suffix to breaker SKU

³ For AFCI/GFCI, add 'DF' suffix to breaker SKU

⁴ For equipment ground fault, add 'GFEP' suffix to breaker SKU

Branch circuit breakers for use with RE panels (480Y/277 V AC 3-phase)

Amp rating	14 kAIC			18 kAIC		
	1-pole	2-pole	3-pole	1-pole	2-pole	3-pole
	Ordering code	Ordering code	Ordering code	Ordering code	Ordering code	Ordering code
15	TEY115	TEY215	TEY315	TEYF115	TEYF215	TEYF315
20	TEY120	TEY220	TEY320	TEYF120	TEYF220	TEYF320
25	TEY125	TEY225	TEY325	TEYF125	TEYF225	TEYF325
30	TEY130	TEY230	TEY330	TEYF130	TEYF230	TEYF330
35	TEY135	TEY235	TEY335	TEYF135	TEYF235	TEYF335
40	TEY140	TEY240	TEY340	TEYF140	TEYF240	TEYF340
45	TEY145	TEY245	TEY345	TEYF145	TEYF245	TEYF345
50	TEY150	TEY250	TEY350	TEYF150	TEYF250	TEYF350
60	TEY160	TEY260	TEY360	TEYF160	TEYF260	TEYF360
70	TEY170	TEY270	TEY370	–	TEYF270	TEYF370
80	TEY180	TEY280	TEY380	–	TEYF280	TEYF380
90	TEY190	TEY290	TEY390	–	TEYF290	TEYF390
100	TEY1100	TEY2100	TEY3100	–	TEYF2100	TEYF3100

ReliaGear® Lighting Panelboards

Branch breakers for RQ panels

THQB and THHQB branch circuit breakers for use with RQ panels (208/120 V AC 3-phase)

Amp rating	10 kAIC			22 kAIC		
	1-pole	2-pole	3-pole	1-pole	2-pole	3-pole
	Ordering code	Ordering code	Ordering code	Ordering code	Ordering code	Ordering code
15	THQB1115 ⁽¹⁾	THQB2115 ⁽¹⁾	THQB32015	THHQB1115	THHQB2115	THHQB32015
20	THQB1120 ^(1,2,3)	THQB2120 ^(1,3)	THQB32020	THHQB1120 ⁽¹⁾	THHQB2120 ⁽¹⁾	THHQB32020
25	THQB1125	THQB2125	THQB32025	THHQB1125	THHQB2125	THHQB32025
30	THQB1130 ^(1,3)	THQB2130 ^(1,3)	THQB32030	THHQB1130	THHQB2130 ⁽¹⁾	THHQB32030
35	THQB1135	THQB2135	THQB32035	–	THHQB2135	THHQB32035
40	THQB1140	THQB2140 ⁽³⁾	THQB32040	THHQB1140	THHQB2140	THHQB32040
45	–	THQB2145	THQB32045	–	–	THHQB32045
50	THQB1150	THQB2150	THQB32050	THHQB1150	THHQB2150	THHQB32050
60	THQB1160	THQB2160	THQB32060	–	THHQB2160	THHQB32060
70	–	THQB2170	THQB32070	–	–	THHQB32070
80	–	THQB2180	THQB32080	–	THHQB2180	THHQB32080
90	–	THQB2190	THQB32090	–	–	THHQB32090
100	–	THQB21100	THQB32100	–	THHQB21100	THHQB32100

⁽¹⁾ For ground fault, add 'GFT' suffix to breaker SKU
⁽²⁾ For AFCI, add 'AF2' suffix to breaker SKU
⁽³⁾ For equipment ground fault, add 'GFEP' suffix to breaker SKU

ReliaGear® Lighting Panelboards

Branch breakers for RE panels

TEY and TEYF branch circuit breakers for use with RE panels (480Y/277 V AC 3-phase)

Amp rating	14 kAIC			18 kAIC		
	1-pole	2-pole	3-pole	1-pole	2-pole	3-pole
	Ordering code	Ordering code	Ordering code	Ordering code	Ordering code	Ordering code
15	TEY115	TEY215	TEY315	–	TEYF215	TEYF315
20	TEY120	TEY220	TEY320	TEYF120	TEYF220	TEYF320
25	TEY125	TEY225	TEY325	–	TEYF225	TEYF325
30	TEY130	TEY230	TEY330	–	TEYF230	TEYF330
35	TEY135	TEY235	TEY335	–	TEYF235	TEYF335
40	TEY140	TEY240	TEY340	–	TEYF240	TEYF340
45	TEY145	TEY245	TEY345	–	TEYF245	TEYF345
50	TEY150	TEY250	TEY350	–	TEYF250	TEYF350
60	TEY160	TEY260	TEY360	–	TEYF260	TEYF360
70	–	TEY270	TEY370	–	TEYF270	TEYF370
80	–	TEY280	TEY380	–	TEYF280	TEYF380
90	–	TEY290	TEY390	–	TEYF290	TEYF390
100	–	TEY2100	TEY3100	–	TEYF2100	TEYF3100

ReliaGear® Merchandised neXT SuperBox Power Panelboards
 SuperBox panels



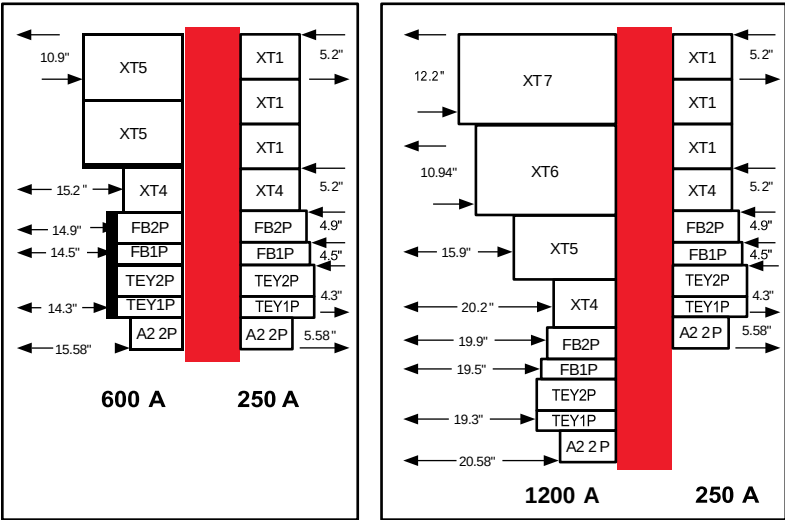
ReliaGear® neXT SuperBox Power Panelboards

Ordering code	Description	Amps	Enclosure type	Main	Dimensions (H x W x D inches)	X-space left	X-space right
RNSB06L6040A	600A MLO NEMA 1 60 x 40 16X no breakers	600 A	NEMA 1	Main lugs (2x #2–600 kcmil)	60 x 40 x 11	12	12
RNSB06B6040A	600A MCB NEMA 1 60 x 40 16X no breakers	600 A	NEMA 1	Main breaker ¹	61 x 40 x 11	16	16
RNSB06L6040R	600A MLO NEMA 3R 60 x 40 16X no breakers	600 A	NEMA 3R	Main lugs (2x #2–600 kcmil)	59 x 40 x 14.5 ²	12	12
RNSB06B6040R	600A MCB NEMA 3R 60 x 40 16X no breakers	600 A	NEMA 3R	Main breaker ¹	60 x 40 x14.5 ²	16	16
RNSB08L7245A	800A MLO NEMA 1 72 x 45 24X no breakers	800 A	NEMA 1	Main lugs (4x #2–600 kcmil)	72 x 45 x 11	20	20
RNSB08B7245A	800A MCB NEMA 1 72 x 45 24X no breakers	800 A	NEMA 1	Main breaker ¹	73 x 45 x 11	24	24
RNSB08L7245R	800A MLO NEMA 3R 72 x 45 24X no breakers	800 A	NEMA 3R	Main lugs (4x #2–600 kcmil)	71 x 45 x 14.5 ²	20	20
RNSB08B7245R	800A MCB NEMA 3R 72 x 45 24X no breakers	800 A	NEMA 3R	Main breaker ¹	72 x 45 x 14.5 ²	24	24
RNSB12L8445A	1200A MLO NEMA 1 84 x 45 32X no breakers	1200 A	NEMA 1	Main lugs (4x #2–600 kcmil)	84 x 45 x 11	28	28
RNSB12B8445A	1200A MCB NEMA 1 84 x 45 32X no breakers	1200 A	NEMA 1	Main breaker ¹	85 x 45 x 11	32	32
RNSB12L8445R	1200A MLO NEMA 3R 84 x 45 32X no breakers	1200 A	NEMA 3R	Main lugs (4x #2–600 kcmil)	83 x 45 x 14.5 ²	28	28
RNSB12B8445R	1200A MCB NEMA 3R 84 x 45 32X no breakers	1200 A	NEMA 3R	Main breaker ¹	84 x 45 x 14.5 ²	32	32

¹ X-space required for main breaker must be accounted for based on specific breaker selected.

² Depth does not include 0.9" of hanger bracket.

Available configurations and wire bend space



40" offset box

45" offset box

All ReliaGear® neXT panelboards are double sided. Wire bending space requirements restrict XT5–XT7 mounting to the left side of the interior.

Wire bend space

Breaker	Left mount space (in)	Right mount space (in)
40" offset enclosure		
XT5	10.9	–
XT4	15.2	5.2
XT1	15.2	5.2
FB 1-pole	14.5	4.5
FB 2-pole	14.9	4.9
TEY 1-pole	14.3	4.3
TEY 2-pole	14.3	4.3
A2 2-pole	15.58	5.58
45" offset enclosure		
XT7	12.2	–
XT6	10.94	–
XT5	15.9	–
XT4	20.2	5.2
XT1	20.2	5.2
FB 1-pole	19.5	4.5
FB 2-pole	19.9	4.9
TEY 1-pole	19.3	4.3
TEY 2-pole	19.3	4.3
A2 2-pole	20.58	5